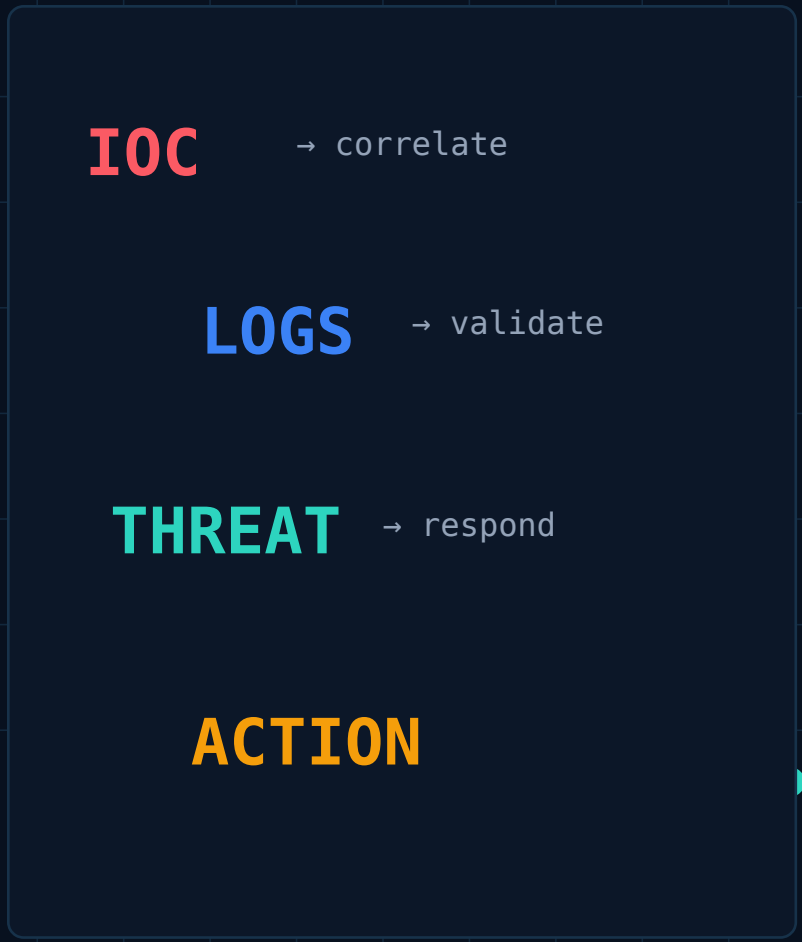
CYB102 GROUP CAPSTONE

SolarWinds IOC Investigation

Detecting suspicious network activity using Splunk and threat intelligence

SPLUNK CORRELATION | VIRUSTOTAL VALIDATION | NIST INCIDENT RESPONSE



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From indicators to actionable incident context

The team correlated known SolarWinds indicators with internal proxy logs, then used external threat intelligence to validate suspicious activity.

2
primary IOC findings highlighted

5
correlated events surfaced in Splunk

HIGH
triage severity assigned

NIST
response framework applied

Objective

Identify internal systems that communicated with known malicious SolarWinds infrastructure and determine the response priority.

What the analysis demonstrated

- Correlation of internal network logs with external IOC data
- Threat validation using VirusTotal vendor detections
- Asset prioritization and remediation using a structured IR process

Scope note: course-provided datasets in a simulated blue-team investigation.

Evidence sources and analysis stack

Two datasets provided the context; Splunk enabled correlation, while VirusTotal provided external threat-intelligence validation.

DATA SOURCE 01

SolarWinds IOC Dataset

Known malicious indicators associated with the SolarWinds campaign.

Key fields

`IP Address · Indicator type · Notes`

DATA SOURCE 02

Network Proxy Logs

Internal network events used to identify which systems communicated with suspicious infrastructure.

Key fields

`Timestamp · IP Address · Computer Name`

ANALYSIS STACK

Splunk

Search, field extraction, and IOC-to-log correlation

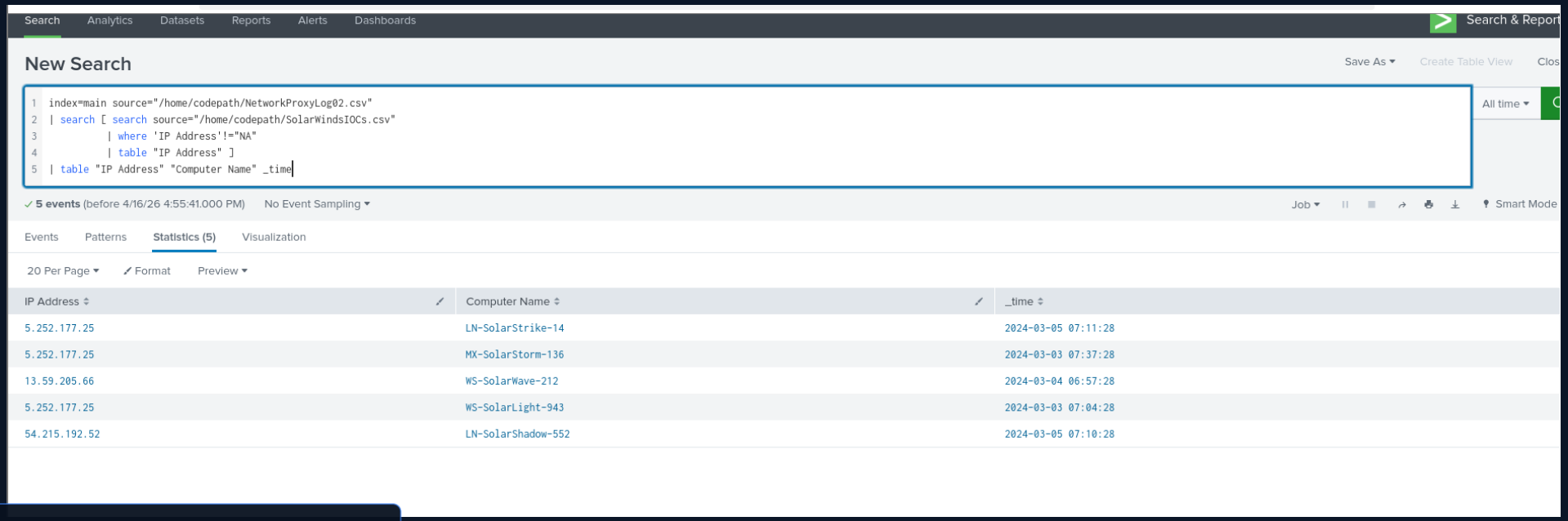
VirusTotal

Independent validation of suspicious IP addresses using multiple security vendors

A repeatable correlation workflow

The analysis moved from raw datasets to validated, prioritized findings using a simple five-step process.

1 **Ingest** → 2 **Extract IOCs** → 3 **Correlate** → 4 **Validate** → 5 **Prioritize**



ORIGINAL SPLUNK EVIDENCE · 5 CORRELATED EVENTS

Query correlates proxy-log IPs with SolarWinds IOC data and returns IP, host, and timestamp for triage.

IOC match: 5.252.177.25

A proxy-log event matched a known SolarWinds indicator and was validated by multiple VirusTotal vendors.

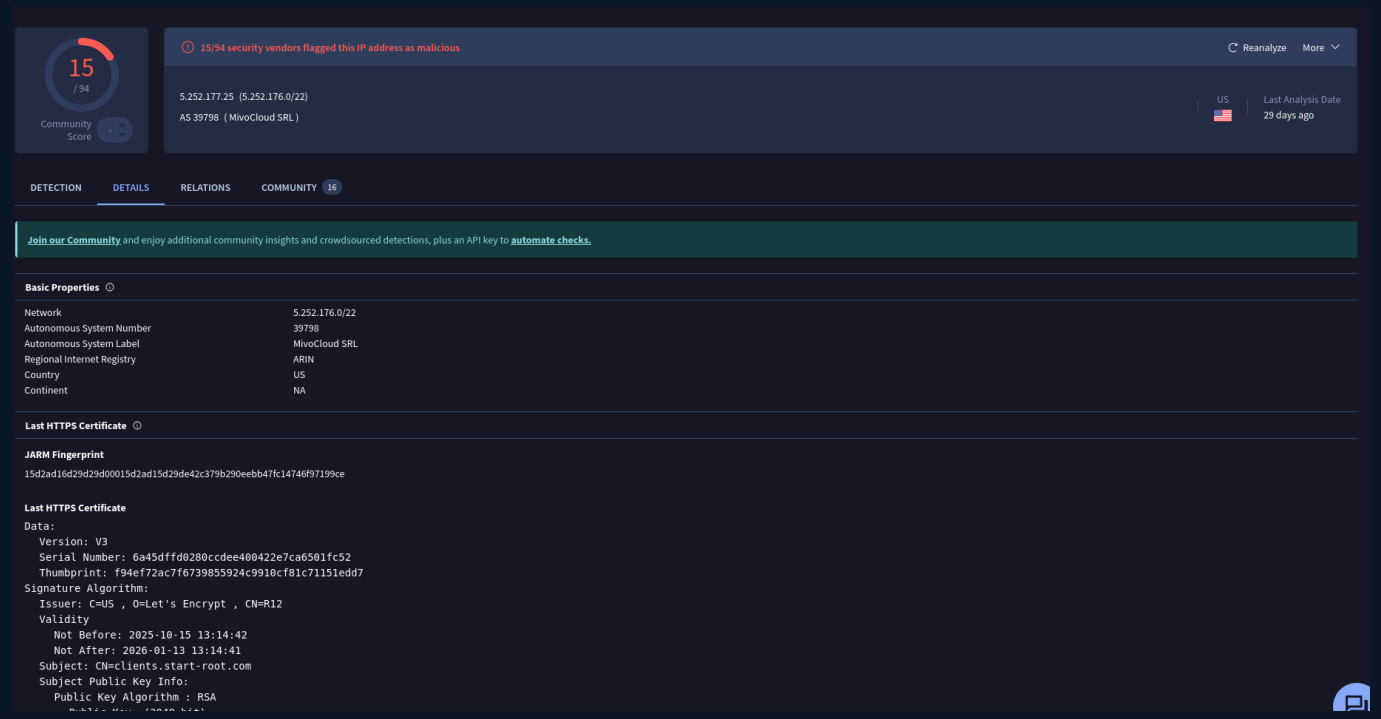
SPLUNK CORRELATION

5.252.177.25

LN-SolarStrike-14

2024-03-05 07:11:28

- Matched in both the internal proxy logs and the IOC dataset
- VirusTotal showed 15 / 94 vendor detections at capture time
- Indicates potential exposure to malicious infrastructure — not proof of compromise



ORIGINAL VIRUSTOTAL EVIDENCE

IOC match: 13.59.205.66

A second host communicated with another IOC; VirusTotal vendor detections reinforced the threat signal.

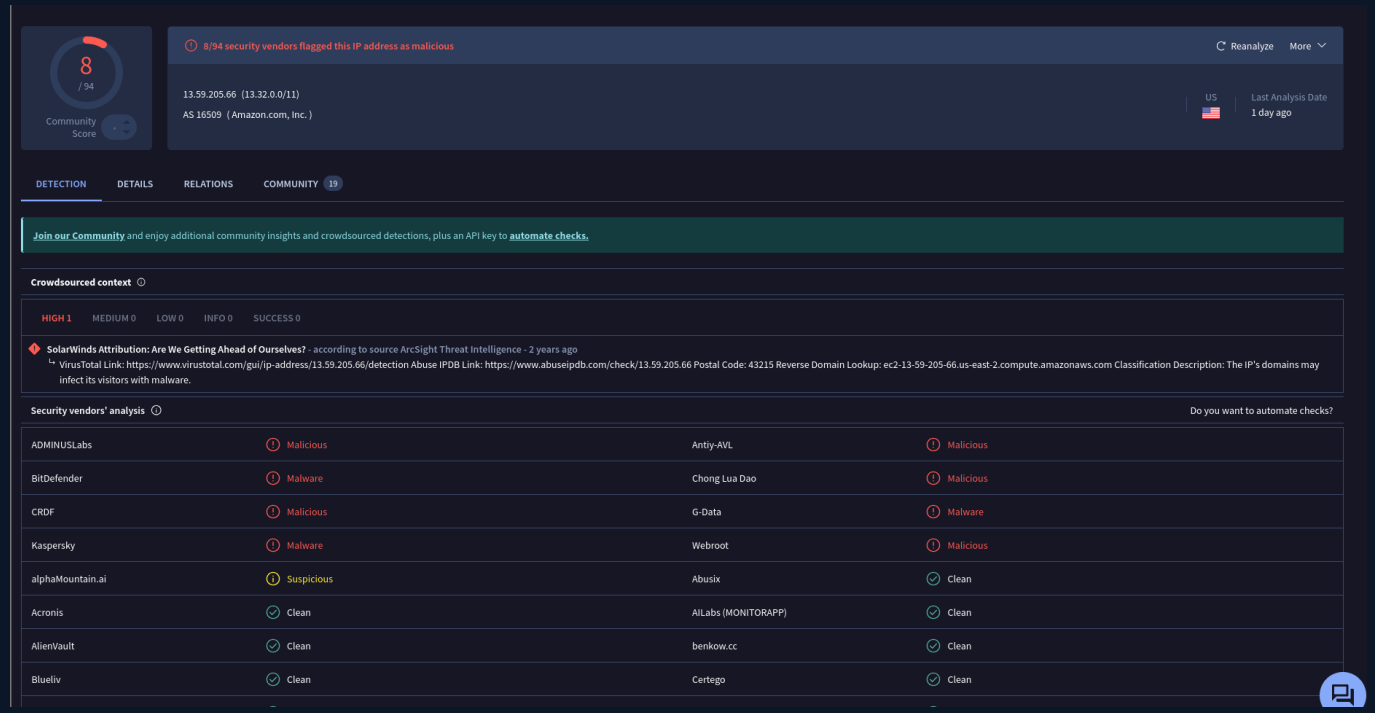
SPLUNK CORRELATION

13.59.205.66

WS-SolarWave-212

2024-03-04 06:57:28

- Matched against the SolarWinds IOC list in Splunk
- VirusTotal showed 8 / 94 vendor detections at capture time
- A second affected system increased the investigation scope and response priority



ORIGINAL VIRUSTOTAL EVIDENCE

Why the incident was prioritized as HIGH severity

The evidence showed repeated communication between internal systems and known malicious infrastructure, requiring immediate containment and deeper investigation.

HIGH

TRIAGE SEVERITY

Potential exposure to known malicious infrastructure across more than one internal system.

Priority: contain, investigate, and monitor.

AFFECTED ASSET 01

LN-SolarStrike-14

Matched 5.252.177.25

AFFECTED ASSET 02

WS-SolarWave-212

Matched 13.59.205.66

Evidence chain

1. **Internal logs**
   Host → external IP
2. **IOC match**
   External IP on SolarWinds list
3. **Threat validation**
   Multiple VT vendors flag malicious
4. **Triage**
   Prioritize containment & investigation

Turn validated indicators into containment actions

Threat intelligence strengthened confidence in the IOC matches; remediation focused on cutting off malicious communication and validating endpoint integrity.

Threat intelligence

- VirusTotal vendor detections supported the SolarWinds IOC matches
- Repeated IOC-to-host correlations increased confidence that the events were not isolated noise
- Vendor counts are point-in-time observations and can change as threat intelligence evolves

Analyst conclusion

Treat the affected hosts as potentially exposed and move quickly to containment and verification.

Recommended remediation

01	**Block malicious IPs**	Firewall / proxy controls
02	**Investigate affected hosts**	Malware scan + endpoint review
03	**Monitor outbound traffic**	Look for repeat connections
04	**Improve detection rules**	Alert on known IOC matches

NIST-aligned response, adapted for IOC-driven log analysis

The team used the NIST incident-response lifecycle as the backbone for a repeatable investigation and case-management process.

01

Preparation

Define sources, tools, and evidence fields

02

Detection & Analysis

Correlate logs with IOC data

03

Containment

Block malicious IP communication

04

Eradication

Investigate / clean affected hosts

05

Recovery

Monitor and strengthen detections

Case-management record

Indicator	Asset	Timestamp	Threat verdict	Response action
5.252.177.25	LN-SolarStrike-14	2024-03-05 07:11:28	**VT: malicious**	Contain + investigate

Repeatable documentation keeps evidence, ownership, and remediation aligned across the response lifecycle.

What this investigation demonstrated

The strongest lesson was not a single IOC — it was the value of combining internal telemetry, external intelligence, and a disciplined response process.

01

Correlation creates context

An IP is more useful when tied to a specific internal host and timestamp.

02

Validation raises confidence

Threat-intelligence sources help distinguish likely malicious infrastructure from background noise.

03

Triage turns data into action

Multiple affected systems and validated IOCs justified a HIGH-priority response.

04

Process makes response repeatable

A NIST-aligned playbook helps teams contain, investigate, recover, and document consistently.

Portfolio takeaway **Splunk + IOC correlation + VirusTotal validation + NIST incident response = a complete blue-team investigation story.**

SolarWinds IOC Investigation

Thank you

Questions?

Team 43

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CYB102 · The Data Dig

